

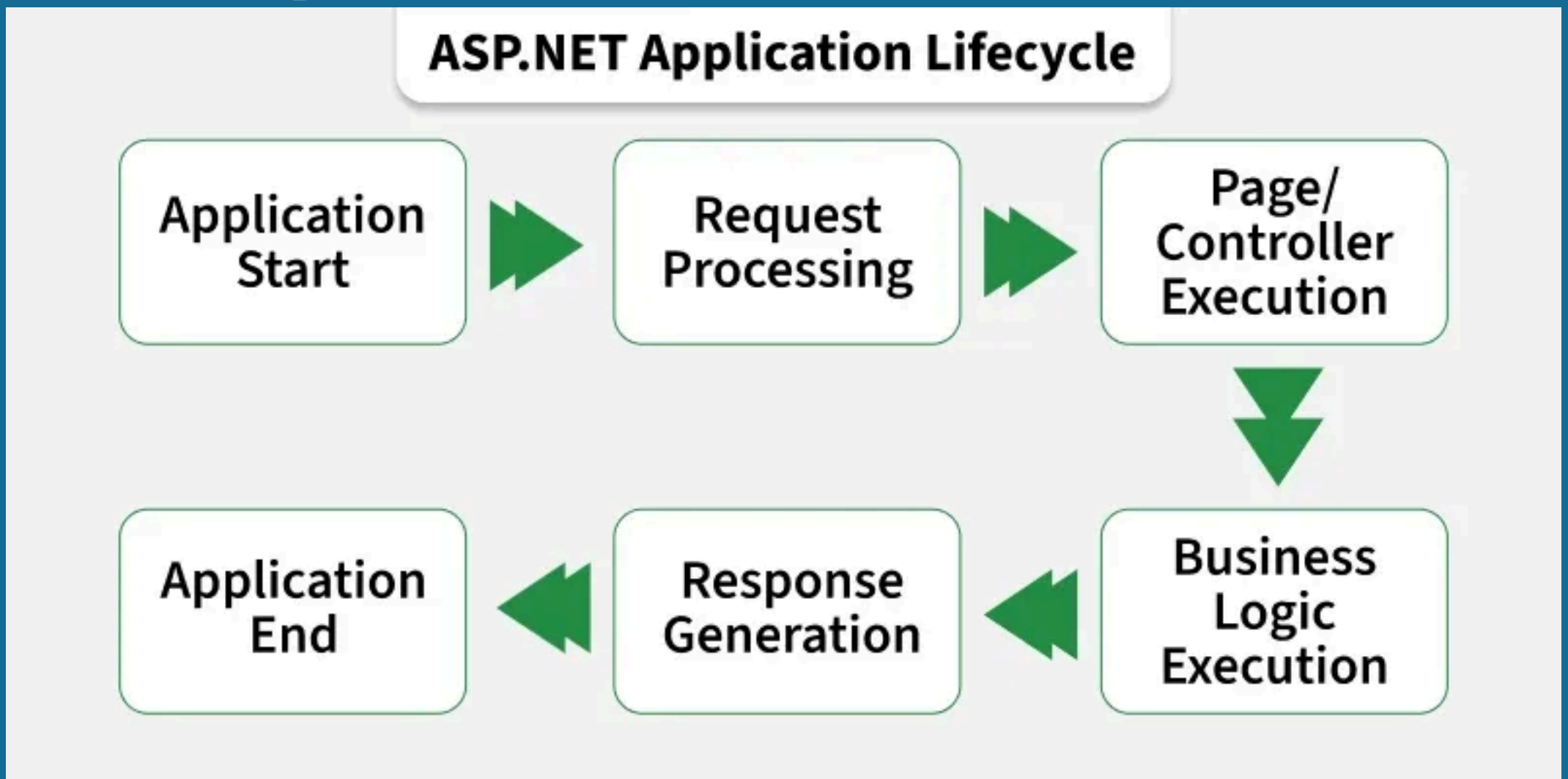
ASP.NET Application Lifecycle - From Client Request to Server Response



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What is the Application Lifecycle? 2



1. **Application Start:** Initialization in Global.asax
2. **Request Processing:** Managed by HttpRuntime
3. **Page/Controller Execution:** Handles events or controller actions
4. **Business Logic Execution:** Validates and processes data, interacts with DAL
5. **Response Generation:** HTML, JSON or XML rendered to client
6. **Application End:** Cleanup of resources



1- Application Start

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- Happens once when the application is first loaded (e.g., IIS starts the app).
- Typical place:
 - Global.asax → Application_Start
- Used for:
 - Initializing routes
 - Dependency injection setup
 - Caching global data
 - Configuring filters, bundles, etc.

📌 This does not run per request—only once per app restart.

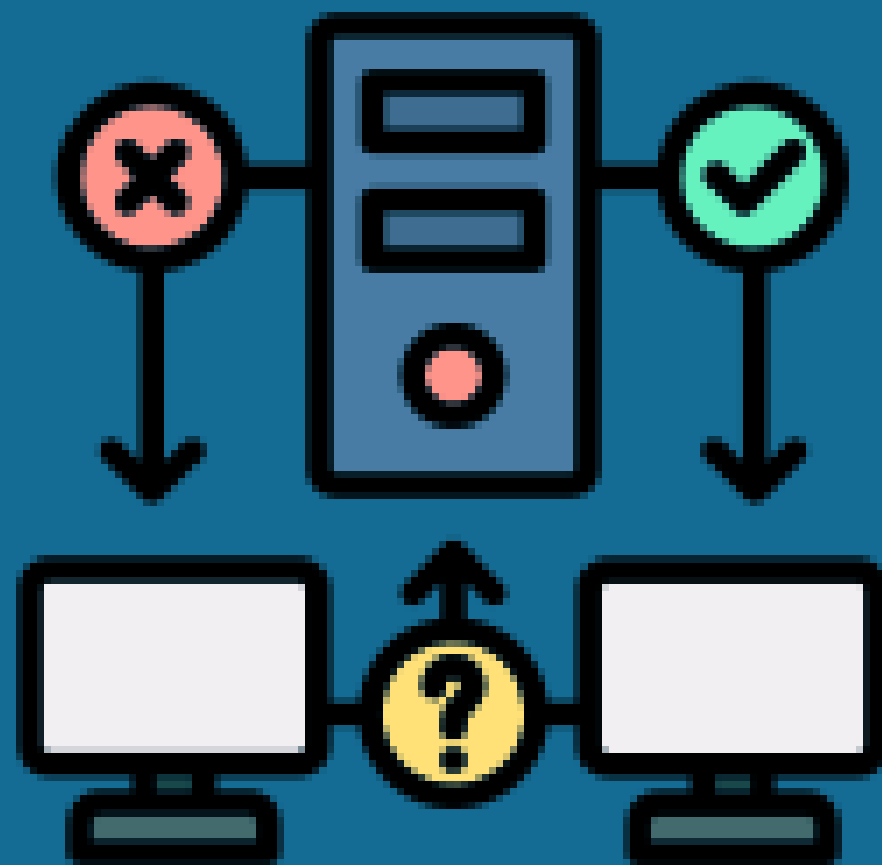


2- Request Processing

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- Triggered for every HTTP request (GET, POST, etc.).
- ASP.NET pipeline starts handling the request.
- Includes:
 - Authentication
 - Authorization
 - Session state
 - HTTP modules & middleware

📌 This is where ASP.NET decides how to handle the request.

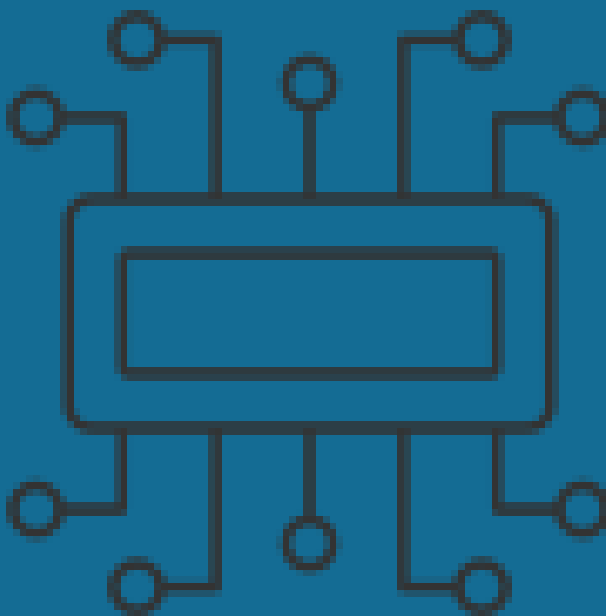


3- Page / Controller Execution

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- ASP.NET routes the request to:
 - Controller (ASP.NET MVC / Web API)
 - Page (Web Forms)
 - Razor Page (ASP.NET Core)
- Examples:
 - HomeController.Index()
 - AccountController.Login()

📌 This is where your endpoint logic starts.

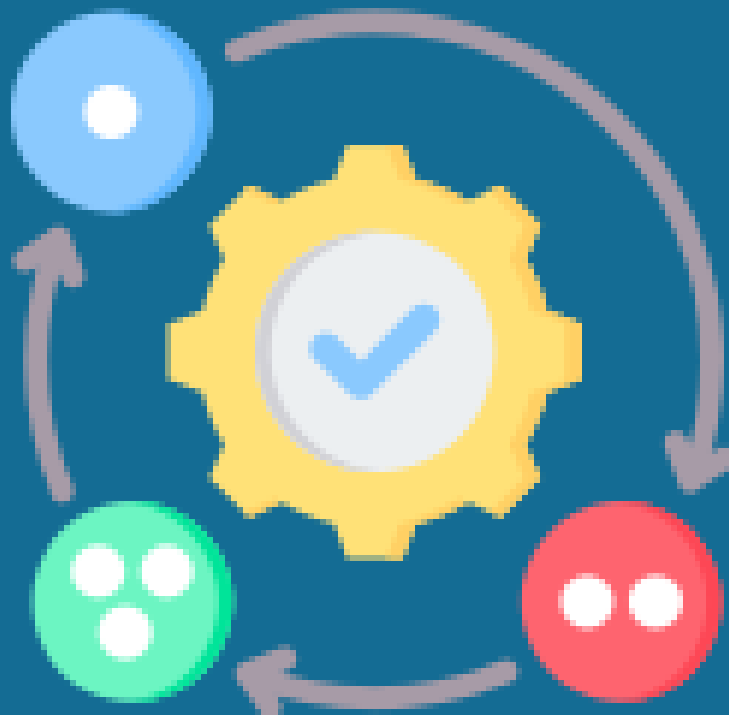


4- Business Logic Execution

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- Core application logic runs here.
- Includes:
 - Service layer
 - Validation
 - Database access
 - External API calls
- Often implemented in:
 - Services
 - Repositories
 - Domain models

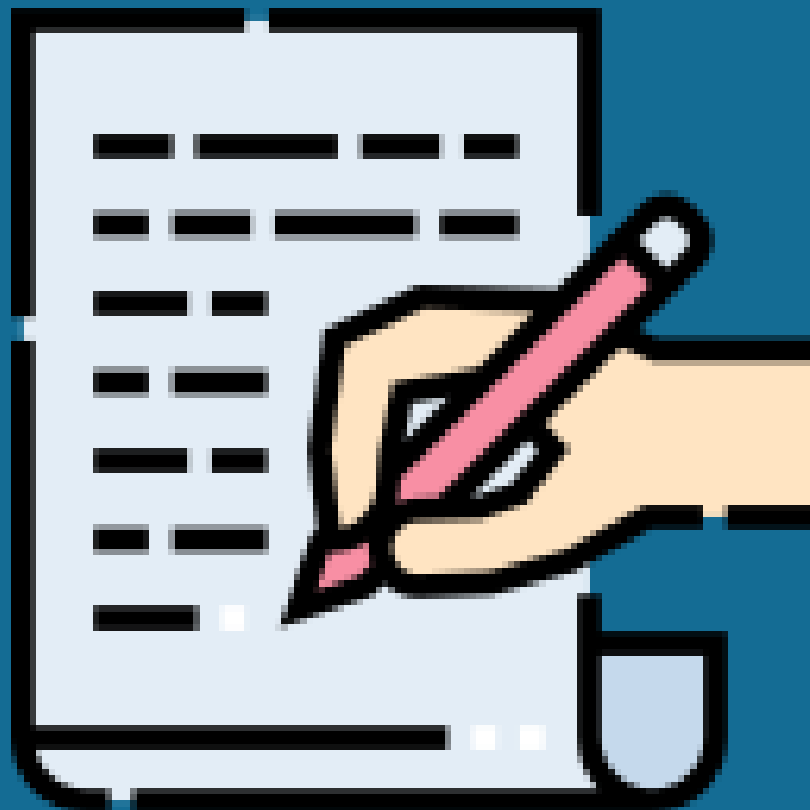
📌 This layer should be independent of UI and HTTP as much as possible.



5- Response Generation⁷

- The result is prepared:
 - HTML (View / Razor)
 - JSON (API response)
 - File download
- Examples:
 - `return View(model);`
 - `return Ok(data);`

📌 ASP.NET converts the result into an HTTP response.

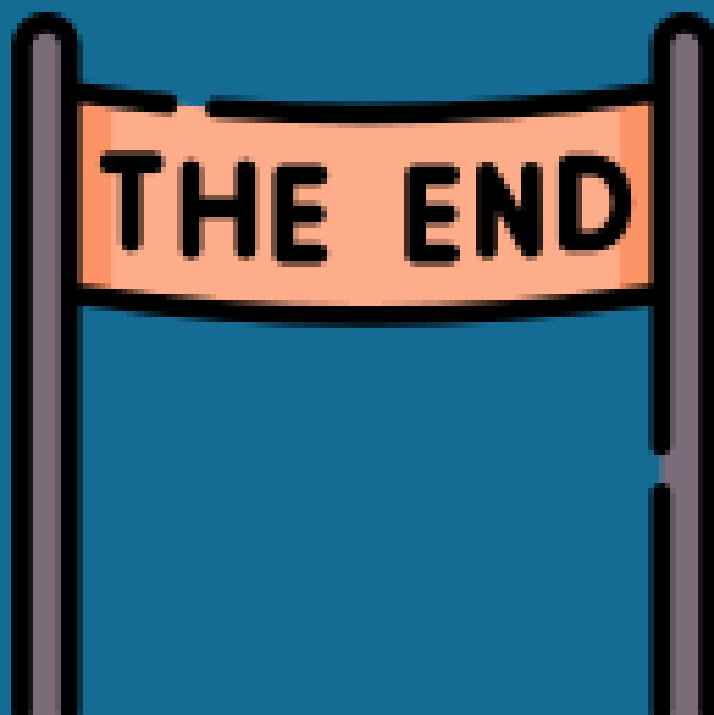


6- Application End

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- Happens when:
 - App pool recycles
 - IIS restarts
 - Web.config changes
- Typical place:
 - Global.asax → Application_End
- Used for:
 - Cleanup
 - Logging
 - Releasing resources

📌 Like Application Start, this runs once, not per request.



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